

Parameter Robustness Improvement of Hybrid Flux Observer for Low Speed Sensorless Drive of PMSM

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Abstract—This letter proposes an improved hybrid flux observer (HFO) for sensorless control scheme of permanent magnet synchronous motor (PMSM). The proposed method is robust against stator resistance mismatch, which is the primary factor affecting the performance of the observer at low speed. A compensation angle has been added to the correction part of the current model and voltage model in HFO, in order to reduce errors in the q-axis direction introduced by stator resistance mismatch. The proposed method modifies the characteristic matrix of the observer, resulting in a larger stable operating speed range and better load-carrying capability. Stability proof and parameter tuning method based on small-signal method is also provided. With only one parameter requiring design, the proposed sensorless scheme is simple to implement. The effectiveness of this method is verified by experiments on a 1.0-kW SPMSM (Surface PMSM) drive platform and a 5.2-kW IPMSM (Interior PMSM) drive platform, demonstrating its good generality.

Index Terms—Sensorless control, PMSM, flux observer, low speed operation, parameter robustness

I. INTRODUCTION

PERMANENT magnet synchronous motor has gained widespread adoption in industrial, transportation, and home appliance areas due to the advantages of high power density and high efficiency. In certain application areas, sensorless technology is required due to cost considerations or environmental constraints [1]. Current location-based technologies can be categorized into signal injection methods (at low speed) and model-based methods (at mid/high speed) [2]. The model-based methods can be further subdivided into flux observer and back-EMF based observe. Unlike EMF, the magnitude of rotor flux amplitude remains constant at low speed. Therefore, the flux observer has the potential for stable operation across the entire speed range [3], especially after the concept of active flux was introduced [4]. Three main problems exist in flux observer at low speed: unobservable rotor position, unstable characteristic matrix, and poor parameter robustness [5].

In terms of observability, a method based on the nonlinear dynamical system state observability theory is proposed, resulting in a unified approach to the ac drive observability analysis [6]. In [7], the local weak observability concept is

used to obtain a unified observable condition. [8] employs virtual voltage injection to achieve zero-speed observability in PMSM. In terms of stability, an adaptive-observer-based active flux observer is proposed, achieving full-speed range operation [9]. In [10], the beneficial effect for of d-axis current on zero-speed and ultra-low-speed stability of flux observer has been demonstrated. In terms of parameter robustness, an adaptive position estimation error suppression method based on a limit cycle oscillator is proposed with parameter robustness [11]. In [12], the dynamic performance of the hybrid flux observer under parameter mismatch has been improved. In [13], a robust observer is proposed and exhibits great parameter robustness under stator resistance variations of $\pm 50\%$. [14] proposes a resistance parameter identification algorithm to avoid resistance mismatch, which is stable near zero frequency. [15] employs the steepest descent optimization to suppress disturbances, demonstrating satisfactory performance under $\pm 20\%$ parameter variations.

This letter proposes an improved hybrid flux observer with compensation angle, which features improves stator resistance parameter robustness and larger stable operating speed range. The proposed sensorless scheme is simple to implement with only one new parameter requiring design, and has provided a reference for it. The effectiveness and generality of the proposed method is verified by experiments on 5.2-kW IPMSM platform and 1.0-kW SPMSM platform respectively.

II. IMPROVED HYBRID FLUX OBSERVER

A. Conventional Hybrid Flux Observer

From the voltage equation of PMSM, the stator flux can be obtained by integrating voltage, which is called voltage model:

$$\psi_s^V = \int (\mathbf{u}_s - R_s \mathbf{i}_s) dt \quad (1)$$

Where $\mathbf{u}_s$ and $\mathbf{i}_s$ are the stator voltage vector and stator current vector in $\alpha\beta$ -axis, R_s is stator resistance, ψ_s is the stator flux. Stator flux can also be obtained from the current, which is called current model:

$$\psi_s^C = P^{-1}(\theta_e) \begin{bmatrix} \psi_d \\ \psi_q \end{bmatrix} = P^{-1}(\theta_e) \begin{bmatrix} \psi_f + L_d i_d \\ L_q i_q \end{bmatrix} \quad (2)$$

Where P^{-1} is the inverse park transform matrix, ψ_d , ψ_q , i_d and i_q are the stator flux and stator current in dq -axis, L_d and L_q are dq -axis inductor.

According to [4], HFO combines the current model and the voltage model by compensation voltage vector which is calculated based on the flux difference between the two models, and it can be expressed as:

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$$\frac{d}{dt} \begin{bmatrix} \hat{\psi}_\alpha^V \\ \hat{\psi}_\beta^V \end{bmatrix} = \begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} - R_s \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \underbrace{K_p \left(\begin{bmatrix} \hat{\psi}_\alpha^C \\ \hat{\psi}_\beta^C \end{bmatrix} - \begin{bmatrix} \hat{\psi}_\alpha^V \\ \hat{\psi}_\beta^V \end{bmatrix} \right)}_{\mathbf{u}_{comp}} \quad (3)$$

K_p is the proportional coefficient of the PI controller, and the integral coefficient is set to 0. The superscript “ $\hat{\cdot}$ ” denotes the observed value.

Based on the concept of active flux, the electric rotor position is given as:

$$\hat{\theta}_e = \arctan \left(\frac{\hat{\psi}_\beta^V - L_q i_\beta}{\hat{\psi}_\alpha^V - L_q i_\alpha} \right) \quad (4)$$

B. Proposed Observer Scheme

Transform $\mathbf{u}_{comp}$ from (3) into observed $\hat{d}\hat{q}$ -axis system.

$$\begin{aligned} K_p P(\hat{\theta}_e) \mathbf{u}_{comp} &= K_p \left(\begin{bmatrix} \psi_f + L_d \hat{i}_d \\ L_q \hat{i}_q \end{bmatrix} - \begin{bmatrix} \hat{\psi}_a + L_q \hat{i}_d \\ L_q \hat{i}_q \end{bmatrix} \right) \\ &= K_p \begin{bmatrix} \psi_f - \hat{\psi}_a + (L_d - L_q) \hat{i}_d \\ 0 \end{bmatrix} \end{aligned} \quad (5)$$

Where $\hat{\psi}_a$ is the observed active flux, and the direction it points to is the observed d -axis direction. From (5), the direction of the compensation vector $\mathbf{u}_{comp}$ aligns with the $\hat{d}$ -axis.

As is shown in Fig. 1, when stator resistance parameter is mismatched, an error vector $(\hat{R}_s - R_s) \mathbf{i}_s$ along $\mathbf{i}_s$ has appeared in the term of $d\psi_s/dt$, which does have a component in $\hat{q}$ -axis, while $\mathbf{u}_{comp}$ has no $\hat{q}$ -axis component.

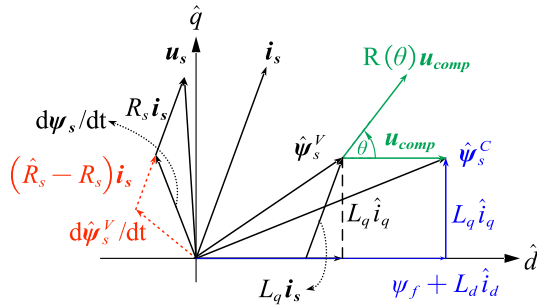


Fig. 1. Vector diagram of error vector in HFO.

Therefore, the proposed observer gives an compensation angle to $\mathbf{u}_{comp}$, so that $\mathbf{u}_{comp}$ has components in both $\hat{d}$ and $\hat{q}$ axes, to better offset errors in the $\hat{q}$ -axis direction. Taking the case where the $\hat{q}$ -axis current is positive with the stator resistance higher than its actual value, the observed angle will lag behind the true angle [13], while the amplitude of the observed flux will decrease. At this point, the positive $\hat{q}$ -axis component of $\mathbf{u}_{comp}$ with the compensation angle drives the observed angle to converge toward the actual value, while its positive $\hat{d}$ -axis component reduces the error in the flux amplitude. With particular consideration on the impact of resistance mismatch, the direction of $\mathbf{u}_{comp}$ is expected to be corrected to the direction of the current $\mathbf{i}_s$. Therefore, the compen-

sation angle should be inversely proportional to $\hat{i}_d$ and directly proportional to $\hat{i}_q$.

The proposed observer is constructed as:

$$\frac{d}{dt} \begin{bmatrix} \hat{\psi}_\alpha^V \\ \hat{\psi}_\beta^V \end{bmatrix} = \begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} - R_s \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + K_p R(\theta) \mathbf{u}_{comp} \quad (6)$$

Where:

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad (7)$$

The diagram of the proposed observer is shown in Fig. 2.

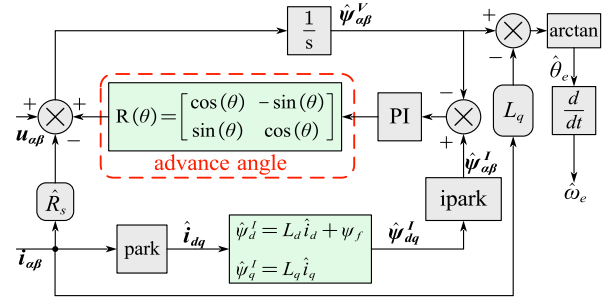


Fig. 2. Control block diagram of the proposed observer.

C. Stability Analysis and Parameter Design

System stability is analyzed using a small-signal model to transform the stability problem of nonlinear system into a linear problem near the equilibrium point. To formulate (6) in terms of a DC component and a disturbance component, it is transformed into the real dq -axis:

$$(p + j\omega_e) \begin{bmatrix} \psi_d^V \\ \psi_q^V \end{bmatrix} = \begin{bmatrix} u_d \\ u_q \end{bmatrix} - R_s \begin{bmatrix} i_d \\ i_q \end{bmatrix} + K_p R(\theta) P(\theta_e) \mathbf{u}_{comp} \quad (8)$$

Ignore the variation of $R(\theta)$, as its rate of change can be constrained within the software, which will be detailed in next section. For (8), consider the small-signal case:

$$\begin{aligned} (p + j\omega_e) \begin{bmatrix} \Delta \psi_d^V \\ \Delta \psi_q^V \end{bmatrix} &= \begin{bmatrix} \Delta u_d \\ \Delta u_q \end{bmatrix} - R_s \begin{bmatrix} \Delta i_d \\ \Delta i_q \end{bmatrix} \\ &+ K_p R(\theta) \left(\begin{bmatrix} \Delta \psi_d^C \\ \Delta \psi_q^C \end{bmatrix} - \begin{bmatrix} \Delta \psi_d^V \\ \Delta \psi_q^V \end{bmatrix} \right) \end{aligned} \quad (9)$$

The flux calculated by the current model in (9) also needs to be transformed into the real dq -axis system from observed $\hat{d}\hat{q}$ -axis:

$$\begin{aligned} \begin{bmatrix} \psi_d^C \\ \psi_q^C \end{bmatrix} &= P(\theta_e) P^{-1}(\hat{\theta}_e) \begin{bmatrix} \hat{\psi}_d^C \\ \hat{\psi}_q^C \end{bmatrix} \\ &= P(\theta_e) P^{-1}(\hat{\theta}_e) \left[\begin{bmatrix} L_d & 0 \\ 0 & L_q \end{bmatrix} P(\hat{\theta}_e) P^{-1}(\theta_e) \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} \psi_f \\ 0 \end{bmatrix} \right] \end{aligned} \quad (10)$$

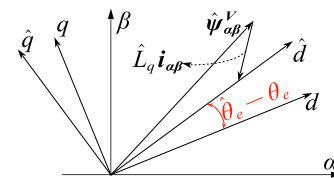


Fig. 3. Relative position of the observed $\hat{d}\hat{q}$ -axis and the real dq -axis.

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From Fig. 3, the angle estimation can be expressed as:

$$\hat{\theta}_e - \theta_e = \arctan\left(\frac{\psi_d^V - L_q i_q}{\psi_q^V - L_q i_d}\right) \quad (11)$$

For (11), consider the small-signal case:

$$\Delta(\hat{\theta}_e - \theta_e) \approx \Delta\left(\frac{\psi_d^V - L_q i_q}{\psi_q^V - L_q i_d}\right) = \frac{\Delta\psi_d^V - L_q \Delta i_q}{\psi_q^V - L_q i_{d0}} \quad (12)$$

Where the subscript “0” denotes the steady-state operating point. Substitute (10)(12) into (9), we can obtain:

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} \Delta\psi_d^V \\ \Delta\psi_q^V \end{bmatrix} &= \begin{bmatrix} -K_P \cos \theta & \omega_e + K_1 \cos \theta \\ -K_P \sin \theta - \omega_e & K_1 \sin \theta \end{bmatrix} \begin{bmatrix} \Delta\psi_d^V \\ \Delta\psi_q^V \end{bmatrix} \\ &+ \begin{bmatrix} L_d K_P \cos \theta - R_s & -L_q K_1 \cos \theta \\ \hat{L}_d K_P \sin \theta & -L_q K_1 \sin \theta - \hat{R}_s \end{bmatrix} \begin{bmatrix} \Delta i_d \\ \Delta i_q \end{bmatrix} + \begin{bmatrix} \Delta u_d \\ \Delta u_q \end{bmatrix} \end{aligned} \quad (13)$$

Where:

$$K_1 = \frac{(L_d - L_q) K_P i_{q0}}{(L_d - L_q) i_{d0} + \psi_f} \quad (14)$$

u_{dq} and i_{dq} represent the input variables of observer, and ψ_{dq} represents the state variable. Then, (13) can be rewritten as:

$$\begin{cases} \Delta \dot{x} = A \Delta x + B \Delta u \\ A = \begin{bmatrix} -K_P \cos \theta & \omega_e + K_1 \cos \theta \\ -K_P \sin \theta - \omega_e & K_1 \sin \theta \end{bmatrix} \end{cases} \quad (15)$$

System stability is analyzed by examining the eigenvalues of matrix A . When all characteristic roots are negative real numbers, the system is stable in non-oscillatory modes. The stable speed range of the observer can be determined:

$$\begin{cases} \omega_e > 0 \\ \omega_e > \omega_{\text{limit}} \end{cases} \quad \text{or} \quad \begin{cases} \omega_e < 0 \\ \omega_e < \omega_{\text{limit}} \end{cases} \quad (16)$$

$$\omega_{\text{limit}} = K_P \left(-\sin \theta + \frac{(L_q - L_d) i_{q0} \cos \theta}{(L_d - L_q) i_{d0} + \psi_f} \right)$$

The compensation angle is constrained by:

$$-\arctan\left(\frac{(L_d - L_q) i_{d0} + \psi_f}{(L_q - L_d) i_{q0}}\right) < \theta < \arctan\left(\frac{(L_d - L_q) i_{d0} + \psi_f}{(L_q - L_d) i_{q0}}\right) \quad (17)$$

From (16), the lowest stable speed is negatively correlated with i_{d0} and positively correlated with i_{q0} . In other words, the heavier the load, the poorer the stability of the PMSM; the greater the current i_d , the stronger the stability of the PMSM.

To maximize the stable speed range, the coefficient θ is given as:

$$\theta = \arctan\left(\frac{(L_q - L_d) i_{q0}}{(L_d - L_q) i_{d0} + \psi_f}\right) \quad (18)$$

Then (16) can be rewritten as:

$$\omega_e \neq 0 \quad (19)$$

This paper improves stator resistance parameter robustness and observer stability of HFO. However, the issue of low-speed unobservability persists, preventing stable operation at extremely low speeds without additional d -axis current [6].

III. DESIGN AND RESULTS OF EXPERIMENT

To verify the effectiveness of the proposed observer method, experiments were performed on the 5.2-kW IPMSM platform and the 1.0-kW SPMSM platform respectively. Motor parameters are listed in TABLE I. A chip TMS320F28075 operating at 120 MHz frequency is applied to execute the sensorless control strategy. The parameter of the hybrid flux observer is set as $K_P=20$. The PWM carrier frequency and the sampling frequency are all set to 8 kHz. The main algorithm of the observer operates at a frequency of 8 kHz, while the compensation angle is calculated at a frequency of 1 kHz.

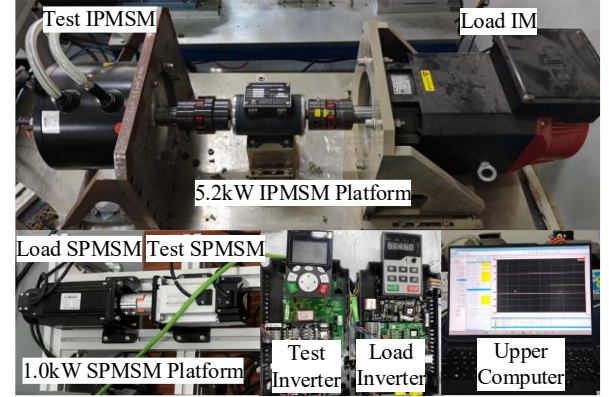


Fig. 4. 5.2-kW IPMSM drive platform and 1.0-kW SPMSM platform

TABLE I
MOTOR PARAMETERS

5.2kW PMSM	Quantity	1.0kW PMSM	Quantity
Pole pairs (p)	3	Pole pairs (p)	5
Rated power [kW]	5.2	Rated power [kW]	1.0
Rated speed [rpm]	2000	Rated speed [rpm]	3000
Rated torque [Nm]	25	Rated torque [Nm]	3.2
Rated current [A]	10.6	Rated current [A]	6.5

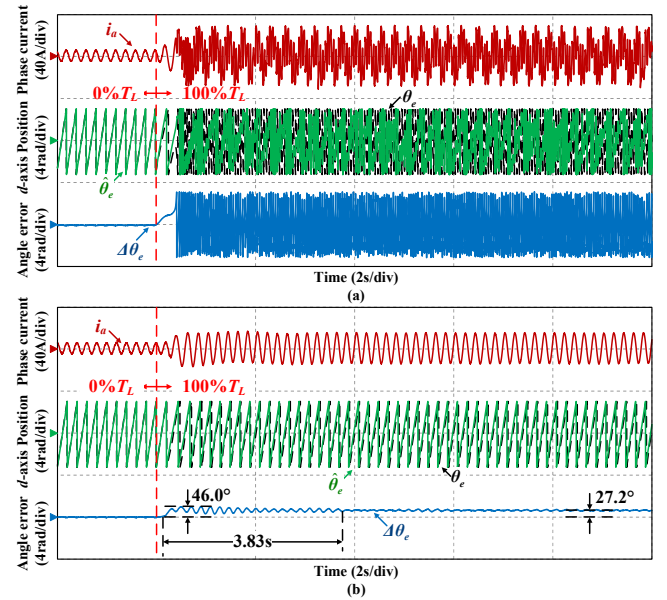


Fig. 5. Experimental results of the 5.2-kW IPMSM when load changes at 5Hz. (a) without compensation angle (b) with compensation angle.

Experimental results of the 5.2-kW IPMSM when load changes from 0 to 100% at 5Hz is shown in Fig. 5, and the d -axis current is set to 0.3 p.u. During the sudden increase in load torque from 0 to 100%, the conventional HFO became

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unstable, and the observed angle exhibited severe oscillations. Due to the inability to output torque effectively, the motor enters a state of reverse rotation. The proposed method exhibits minor oscillations in the observed angle after a load transient (with a maximum angle error of 46.0°), yet remains stable. For coupling relationship between the compensation angle and i_{q0} , the compensation angle requires a convergence process after the load change. After stabilization, the angle error is 27.2° , and the system maintains the ability to steadily deliver the rated load.

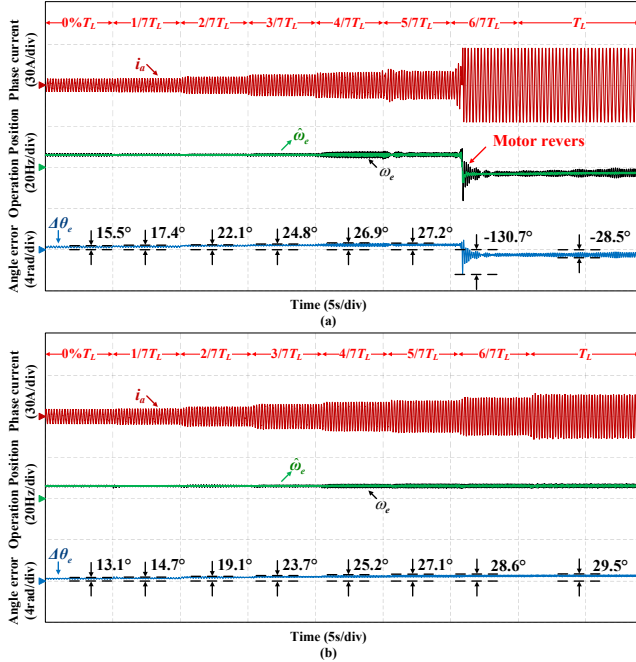


Fig. 6. Experimental results of the 5.2-kW IPMSM load test at 5Hz when $\Delta R_s = +100\%$. (a) without compensation angle (b) with compensation angle.

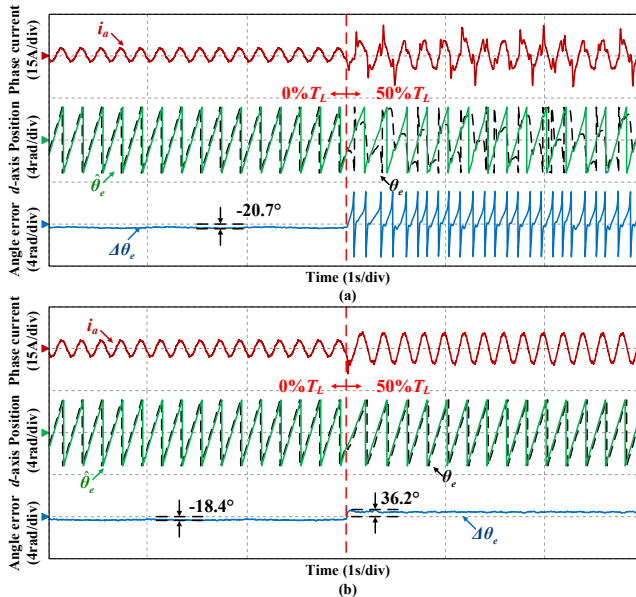


Fig. 7. Experimental results of the 1.0-kW SPMSM load test at 5 Hz when $\Delta R_s = +100\%$. (a) without compensation angle (b) with compensation angle.

As is shown in Fig. 6, the robustness of the proposed method against stator resistance mismatch was compared with the conventional HFO from the 5.2-Kw PMSM. Under the test

conditions—where the stator resistance was set to 200% of its actual value and the d-axis current was set at 0.3 p.u.—the load torque was increased stepwise from 0 to 100% at intervals of approximately 5 s. The conventional HFO exhibits distortion in the estimated angle once the load torque reaches 6/7 rated value, leading to motor reversal. In contrast, the proposed method maintains accurate position estimation and stable operation throughout the entire loading process.

Fig. 7 shows the experimental results of the 1.0-kW PMSM when load changes from 0 to 50% at 5Hz, with stator resistance set as 200% of actual value. The d-axis current is set to 0.25 p.u. Before loading, the angle error of the proposed method is -18.4° , whereas the conventional method is -20.7° . After loading, the conventional HFO diverges, while the proposed method remains stable with a maximum angle error of 36.2° .

IV. CONCLUSION

This manuscript presents a novel method for parameter robustness and low speed stability improvement for PMSM sensorless control. The suppression of stator resistance mismatch by the proposed method is illustrated using a geometric approach. Based on small signal analysis, the proposed method modifies the characteristic matrix of the observer, resulting in a lower stable operating speed. To verify the generality of the proposed method, experiments are conducted on both IPMSM and SPMSM. Compared with the traditional HFO, the proposed method has a better suppression ability for the stability degradation caused by stator resistance parameter mismatch, and exhibits smaller estimation error across varying load conditions and under parameter mismatches.

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